

Towards Robust Few-Shot Object Detection: Attribute Semantic Manifold Modeling and Rectification

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Abstract

Few-shot object detection (FSOD) aims to recognize novel categories with extremely limited annotated samples. Despite the success of the two-stage fine-tuning paradigm in FSOD, classification discriminability remains a key bottleneck. In this paper, we identify two coupled challenges that exacerbate this bottleneck: 1) semantic drift, where novel class representations drift toward similar base classes due to biased category representations under scarce supervision; and 2) negative bias, where unlabeled foreground objects in the background give incorrect negative gradients that distort decision boundaries. To address these issues, we propose an Attribute Semantic Manifold Modeling and Rectification (ASMR) method for FSOD. Specifically, we design an Attribute Semantic Manifold Modeling (ASM) module, which grounds category representations in a structured attribute semantic manifold. By leveraging Large Language Models (LLMs) and CLIP to generate and encode fine-grained attributes, ASM models attribute-category relations via a hypergraph and aligns visual RoI features with this manifold to mitigate semantic drift. Meanwhile, we propose an Attribute-Guided Background Rectification (AGR) module to alleviate negative bias. Exploiting the early semantic recognition capability of the attribute semantic manifold, AGR identifies potential foreground objects in the background to suppress incorrect negative gradients and recovers missing supervision via online pseudo-labeling. ASM and AGR operate synergistically to enable more stable semantic alignment. Experimental results on the PASCAL VOC and MS COCO benchmarks demonstrate that ASMR achieves competitive performance, particularly in extreme low-shot regimes, effectively improving the discriminative capability of few-shot detectors. Code will be made publicly available.

CCS Concepts

• Computing methodologies → Object detection.

Keywords

Few-Shot Object Detection, Attribute Semantic Manifold

1 Introduction

Few-shot object detection (FSOD) aims to learn and recognize novel categories from only a few annotated samples. It typically follows the two-stage paradigm of base class pretraining and novel class fine-tuning [11, 19, 40]. Under extreme data constraints, FSOD remains highly sensitive to class similarity and insufficient annotations [30, 31]. Although existing methods improve detection performance through some strategies (such as classifier decoupling, contrastive learning, and prototype calibration) [25, 35, 39], the classification discriminability after novel class fine-tuning remains a key bottleneck.

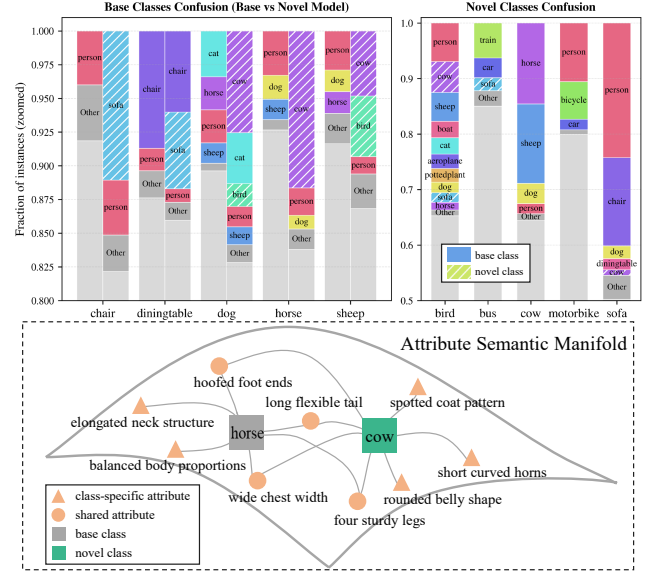


Figure 1: Error analysis and attribute semantic manifold illustration. Top: selected statistics of classification error sources for the MFDC [42] model under the 1-shot setting on PASCAL VOC Split 1 with fixed ground truth bounding boxes, showing that confusion is mainly concentrated in semantically similar base/novel class pairs. Bottom: illustration of the proposed attribute semantic manifold, where shared attributes facilitate transfer across related classes and class-specific attributes improve class discriminability.

To better understand this bottleneck, we perform classification on regions of interest (RoIs) using a representative FSOD method, MFDC [42] with ground-truth bounding boxes fixed, thereby removing the influence of localization errors. As shown in the top panel of Fig. 1, even under ideal localization conditions, classification predictions still exhibit significant confusion [25, 30, 39], where classification errors mainly arise from semantically similar base/novel classes. This empirical result demonstrates that insufficient discriminative capability between semantically similar classes constitutes a fundamental challenge in FSOD.

On the one hand, under extremely limited supervision, existing methods struggle to learn discriminative class-specific representations from very limited annotated samples. This often results in biased category representations [28, 45, 50], causing novel class representations to drift toward semantically similar base classes, a phenomenon we refer to as **semantic drift**. Moreover, these methods rely solely on visual information [39] and lack explicit

semantic guidance to model the fine-grained relationships among semantically similar categories [50].

On the other hand, existing methods primarily focus on exploiting positive samples [41], while overlooking the effect of **negative bias** (i.e., the interference caused by unlabeled foreground objects in the background) [6]. As pointed out in [14, 35], training images in FSOD settings often contain multiple objects but only a few of them are annotated. As a result, many unlabeled foreground objects (either from base or novel classes) are inevitably treated as background during training. While the labeled positive RoIs provide reliable supervision, a portion of sampled negative RoIs are false negatives caused by missing annotations [17, 29]. These false negatives bias the classifier to incorrectly identify true foreground objects as background, which further distorts the decision boundary and exacerbates the representation drift of novel classes.

To address these issues, we propose an Attribute Semantic Manifold Modeling and Rectification (ASMR) method for FSOD. Our method is motivated by the observation that when category representations are biased under limited visual supervision, fine-grained semantic attributes (such as ‘a long flexible tail’ for both horse and cow) provide a more transferable and reliable basis that can be shared across related categories [1]. Hence, instead of representing each category as a single isolated prototype derived from limited observations, we hypothesize that category representations reside on a structured, low-dimensional manifold—termed the attribute semantic manifold—where attributes serve as the basis for category representations. As illustrated in the bottom panel of Fig. 1, this manifold is structured by combinations of shared and class-specific attributes, and thus serves as a geometric prior for regularizing few-shot category representations.

Accordingly, meaningful class representations reside within a low-dimensional subset of the ambient embedding space [2, 16, 38], where the intrinsic geometry is governed by the composition of semantic attributes. By aligning novel classes with this structured manifold, the model becomes more robust to biased category representations and effectively mitigates semantic drift toward similar base categories. Moreover, the attribute semantic manifold provides robust semantic priors for unlabeled foreground objects during fine-tuning (even before the visual classifier becomes reliable). This characteristic provides a reliable and stable cue for identifying potential foreground objects (i.e., unlabeled foreground objects) among background RoIs [17, 21], helping reduce negative bias.

Specifically, we design an Attribute Semantic Manifold Modeling (ASM) module that constructs a structured attribute semantic manifold and aligns visual RoI features with it to mitigate semantic drift. ASM first uses a Large Language Model (LLM) to generate textual fine-grained attributes for each category, which are then encoded into the CLIP text embedding space [4, 36]. To reduce redundancy, semantically similar fine-grained attributes are merged into a compact attribute set. Their many-to-many relations with categories are naturally modeled as a hypergraph [12], where attributes and categories are treated as vertices and hyperedges, respectively. Category prototypes are then represented as weighted combinations of attributes, where the weights can be interpreted as attribute-based coordinates in the attribute semantic manifold. During training, ASM further guides representation learning by aligning RoI features with this attribute semantic manifold. In this way, semantic priors

are incorporated into representation learning as structured constraints, leading to more robust category prototypes and improved discrimination among semantically similar classes.

Moreover, we design an Attribute-Guided Background Rectification (AGR) module to reduce negative bias. Traditional methods for handling negative samples often rely on visual classifiers. This makes them prone to confirmation bias [29], where unlabeled foreground objects are repeatedly penalized as background. In contrast, AGR exploits the early semantic recognition capability of the attribute semantic manifold during fine-tuning. In particular, AGR estimates attribute-guided confidences to identify background RoIs and suppress incorrect negative gradients. For samples satisfying higher confidence thresholds, AGR further performs confidence-aware supervision rectification to recover potential supervision. In this way, AGR rectifies the biased optimization process without introducing additional penalties.

Importantly, ASM and AGR operate synergistically. The attribute semantic manifold constructed by ASM provides reliable semantic priors that help AGR filter out false negatives in the early stage of fine-tuning, while the rectified supervision from AGR in turn enables ASM to learn more stable semantic alignment later on.

The main contributions of this paper are summarized as follows:

- We propose the ASM module to mitigate semantic drift, which grounds few-shot category representations in an attribute semantic manifold, thereby providing transferable semantic priors.
- We propose the AGR module to alleviate negative bias caused by missing annotations by identifying potential foreground objects among background RoIs, suppressing incorrect negative gradients, and recovering missing supervision for high-confidence samples.
- Experimental results show that ASMR achieves competitive performance in extreme low-shot regimes, effectively improving the discriminative capability of few-shot detectors.

2 Related Work

2.1 Few-Shot Object Detection

Early FSOD methods are predominantly based on the meta-learning paradigm [44]. Pioneering works (such as FSRW [22]) employ episodic training and feature reweighting to transfer meta-level knowledge from base to novel classes. Attention-RPN [10] improves support-query matching through class-attentive proposal generation and multi-relation reasoning. More recently, fine-tuning-based methods [5, 35, 40, 46, 49] have shown strong performance and become a widely adopted paradigm. TFA [40] fine-tunes only the detection head, whereas FSCE [39] and DeFRCN [35] respectively improve adaptation through supervised contrastive learning on proposal features and decoupled optimization with prototypical calibration. MFDC [42] explicitly distills multi-faceted commonalities between base and novel classes through a unified memory-bank-based framework. In parallel, metric-learning-based methods impose more informative structure on the embedding space. Rep-Met [23] adopts representative-based metric learning with multiple learned class representatives. KFSOD [47] introduces kernelized region representations with efficient integral aggregation to model

non-linear similarities. Despite their success, these methods still rely only on visual cues derived from limited annotated samples.

Recently, several studies have observed the issue of negative bias, where unlabeled foreground objects are mistakenly treated as background during fine-tuning. The negative bias can introduce incorrect gradients and thus impair representation learning. To reduce this, prior works typically rely on gradient decoupling strategies [14, 35] or heuristic background filtering mechanisms [26] to isolate or discard false negative samples. However, these methods still rely mainly on visual cues, which are often unstable at the early stage of fine-tuning.

Different from the above methods, our method addresses both semantic drift and negative bias under a unified manifold learning paradigm. Instead of constructing category representations purely from limited visual samples, we reconceptualize prototype learning by anchoring it within an attribute semantic manifold, which provides structured and transferable semantic priors. Informed by this manifold, we perform attribute-guided background rectification to enable more reliable identification of unlabeled foreground objects. By establishing a functional synergy between semantic modeling and background rectification, our method offers a more robust learning objective for few-shot settings.

2.2 Semantic Priors in Object Detection

The use of semantic priors to facilitate knowledge transfer is pioneered in zero-shot learning (ZSL) [13, 34], where categories are represented by shared semantic descriptors or attributes [24]. In object detection, this line of research has evolved into zero-shot [1] and open-vocabulary [17] settings.

Despite the great progress, existing semantic-enhanced object detection methods often treat semantic priors as isolated points in a high-dimensional space. For instance, category embeddings derived from CLIP or Word2Vec are often utilized as independent anchors [1, 17], overlooking the structured relationships among category semantics (such as shared attributes and higher-order correlations). While some works [7, 8, 50] incorporate knowledge graphs to model inter-class dependencies, they are often constrained by pre-defined relations or hierarchical structures.

Moreover, according to the manifold hypothesis [2, 3, 16], high-dimensional data tend to lie on or near a structured low-dimensional manifold. Rather than treating class representations as isolated points, a manifold-based perspective suggests that explicitly modeling the structure of the underlying manifold can provide stronger inductive biases for few-shot recognition. However, effectively constructing such a semantic manifold from textual descriptions and attribute semantics remains challenging [33].

Unlike existing methods, our method departs from the conventional point-wise modeling of semantic priors and instead explicitly constructs a structured attribute semantic manifold. Rather than directly using category-level embeddings as fixed anchors, we decompose category semantics into fine-grained attribute components and model their interactions through hypergraph propagation. This enables the learned representation to capture both shared and class-specific semantics in a data-driven manner, without relying on manually defined hierarchies or external knowledge graphs.

3 Method

3.1 Problem Settings and Overview

We follow the standard FSOD settings [22, 40]. Let C_{base} and C_{novel} denote the sets of base and novel categories, respectively, with $C_{base} \cap C_{novel} = \emptyset$. Training is conducted over two datasets in two stages (pretraining on the base training set $\mathcal{D}_{base}$ and fine-tuning on the novel training set $\mathcal{D}_{novel}$). $\mathcal{D}_{base}$ contains abundant bounding-box annotations for categories in C_{base} , and is used to learn transferable detection representations. $\mathcal{D}_{novel}$ contains only K annotated instances per category for C_{novel} , where $K \in \{1, 2, 3, 5, 10, 30\}$, and is used for few-shot adaptation to novel categories.

Notably, while the category labels of the two stages are disjoint, the image content is not strictly separated at the instance level. Novel category objects may appear in $\mathcal{D}_{base}$ without annotation, and images in $\mathcal{D}_{novel}$ may also contain additional unlabeled novel instances serving as background samples. This realistic setting aligns with the fundamental motivation of FSOD: learning transferable representations from richly annotated base categories and rapidly adapting them to novel categories with extremely limited supervision, while reflecting the realistic challenge of incomplete instance annotation.

In this paper, we propose an Attribute Semantic Manifold Modeling and Rectification (ASMR) method. Based on MFDC [42], ASMR introduces two key modules: an Attribute Semantic Manifold Modeling (ASM) module and an Attribute-Guided Background Rectification (AGR) module. ASM is designed to reduce biased category representations with our proposed attribute semantic manifold. Instead of estimating category prototypes only from a few visual samples, we construct them from external priors and align RoI features to this semantic manifold. AGR is designed to reduce negative bias introduced by missing annotations during novel class fine-tuning. In particular, we exploit the early semantic recognition capability of the attribute semantic manifold to identify potential foreground objects in the background.

The overview of our method is given in Fig. 2. Given an input image, the backbone and region proposal network (RPN) produce region proposals. After RoI pooling, the i -th RoI is represented by a visual feature $\mathbf{f}_i \in \mathbb{R}^{d_f}$, where d_f denotes the feature dimension. We process $\mathbf{f}_i$ with two weakly-coupled branches: (1) the detection branch with a standard Faster R-CNN style head for box regression and visual classification, which contains the same commonalities distribution design as MFDC [42]. (2) the attribute branch (consisting of the ASM and AGR modules) which maps RoI features into an attribute semantic manifold to perform semantic alignment and label rectification. The attribute branch receives RoI features through a gradient scaling operator, so that semantic supervision assists representation learning without destabilizing the detector backbone during early fine-tuning.

3.2 Attribute Semantic Manifold Modeling

The ASM module consists of two stages: attribute semantic manifold construction (ASMC) and visual-semantic alignment. ASMC first constructs an attribute semantic manifold, and the subsequent alignment stage aligns visual RoI features with this manifold to mitigate semantic drift.

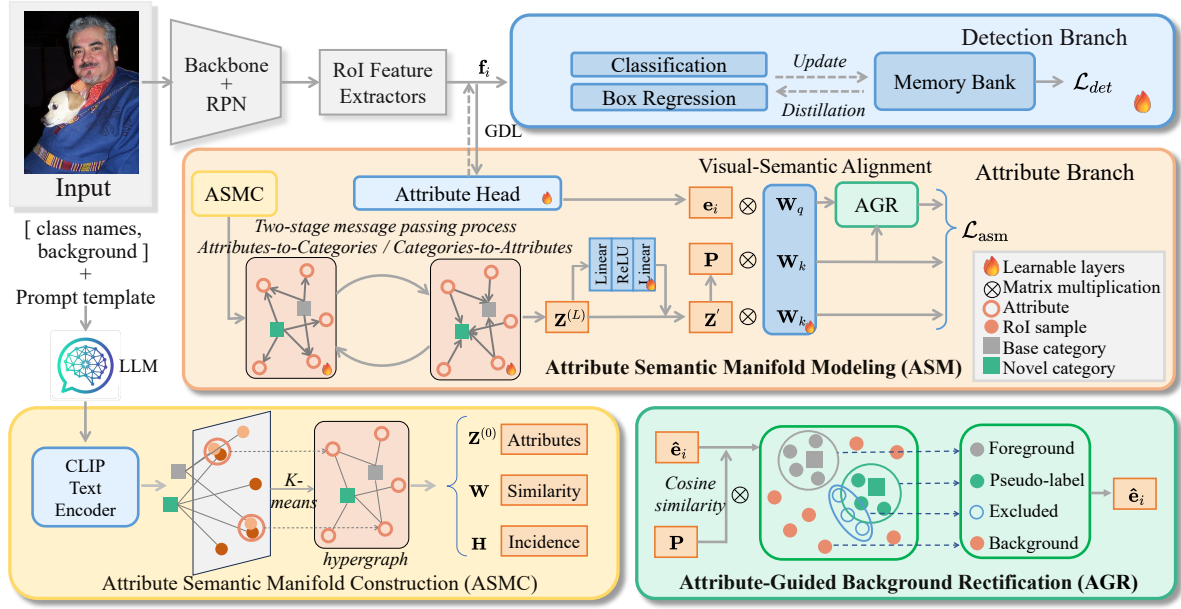


Figure 2: Overview of the proposed method. An input image is processed into RoI features f_i , which are fed into a detection branch and an attribute branch (containing two modules: ASM and AGR). ASM first constructs an attribute semantic manifold from LLM- and CLIP-derived attributes, yielding the initial attribute matrix $Z^{(0)}$, similarity matrix W , and incidence matrix H . It then refines the attributes into Z' and obtains category prototypes P for visual-semantic alignment. AGR then uses the aligned embeddings and prototypes to rectify background RoIs into pseudo-labeled foreground, excluded samples, or reliable background samples. Before the attribute head, GDL attenuates gradients from the attribute branch to stabilize optimization.

3.2.1 Attribute Semantic Manifold Construction. We consider all foreground classes $c \in C_{base} \cup C_{novel}$ together with the background class, yielding a total of $C + 1$ categories, where $C = |C_{base} \cup C_{novel}|$. For each category, we use a set of predefined prompt templates to query a Large Language Model (LLM), such as ChatGPT [4, 32], to generate fine-grained textual attribute descriptions. Each description is then encoded by a pre-trained CLIP text encoder into an embedding vector. To reduce redundant or semantically overlapping descriptions, we cluster the resulting embeddings using K -means on the normalized embeddings, where cosine similarity serves as the semantic distance measure. Let z_n denote the n -th attribute. The initial attribute matrix is

$$Z^{(0)} = [z_1^{(0)}, \dots, z_{N_a}^{(0)}]^\top \in \mathbb{R}^{N_a \times d_a}. \quad (1)$$

Here, N_a denotes the number of attributes, and d_a is the attribute dimension.

Then, we build an attribute-category incidence matrix $H \in \{0, 1\}^{N_a \times (C+1)}$ based on the clustered attributes and their category affiliations, where $H_{i,j} = 1$ indicates that the i -th attribute is associated with the j -th category, and $H_{i,j} = 0$ otherwise.

To capture high-order semantic dependencies among attributes, we model attributes as vertices and categories as hyperedges, and perform hypergraph propagation [12] over the incidence matrix H . This propagation follows a two-stage message passing process, namely vertex-to-hyperedge aggregation and hyperedge-to-vertex redistribution, which can be written in matrix form as

$$A_h = D_v^{-1/2} H D_e^{-1} H^\top D_v^{-1/2}, \quad (2)$$

where $D_v \in \mathbb{R}^{N_a \times N_a}$ and $D_e \in \mathbb{R}^{(C+1) \times (C+1)}$ are the diagonal vertex- and hyperedge-degree matrices, respectively, with $(D_v)_{n,n} = \sum_{c=1}^{C+1} H_{n,c}$ and $(D_e)_{c,c} = \sum_{n=1}^{N_a} H_{n,c}$. The matrix A_h captures vertex-hyperedge-vertex propagation, so that each attribute aggregates information from others that co-occur within the same categories.

To further encode direct pairwise semantic relations between attributes, we introduce a cosine-similarity matrix $W \in \mathbb{R}^{N_a \times N_a}$ computed from the attributes in $Z^{(0)}$. Combining high-order hypergraph structure, pairwise attribute similarity, and self-connections, the final propagation operator is defined as

$$A = \text{RowNorm}(A_h + \alpha W + I), \quad (3)$$

where $\text{RowNorm}(\cdot)$ denotes row-wise ℓ_1 normalization, $I \in \mathbb{R}^{N_a \times N_a}$ is the identity matrix, and $\alpha \in \mathbb{R}$ is a scalar coefficient.

Let $Z^{(l)} \in \mathbb{R}^{N_a \times d_a}$ be the attribute matrix at layer l , where each row corresponds to one attribute. The layer-wise propagation is given by

$$Z^{(l+1)} = \text{Norm}\left(Z^{(l)} + \Theta^{(l)}(AZ^{(l)})\right), \quad (4)$$

where $\Theta^{(l)}(\cdot)$ is a lightweight MLP consisting of two linear layers with a ReLU activation in between and $\text{Norm}(\cdot)$ denotes row-wise ℓ_2 normalization.

Since the attributes are derived from the semantic embedding space of the CLIP text encoder, we further introduce a residual mapping $\psi(\cdot)$ to adapt these semantic attribute representations to the target visual distribution during training. For each attribute

$\mathbf{z}_n^{(L)} \in \mathbf{Z}^{(L)}$, the refined attribute is defined as

$$\mathbf{z}'_n = \text{Norm}\left(\mathbf{z}_n^{(L)} + \psi(\mathbf{z}_n^{(L)})\right), \quad n \in \{1, \dots, N_a\}, \quad (5)$$

where L denotes the final layer in hypergraph propagation. Let $\mathbf{Z}' \in \mathbb{R}^{N_a \times d_a}$ denote the resulting refined attribute matrix. We then obtain category prototypes by aggregating the associated attributes according to the incidence matrix $\mathbf{H}$. In matrix form, all category prototypes are given by

$$\mathbf{P} = \text{Norm}(\mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{Z}'), \quad (6)$$

where $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_{C+1}]^\top \in \mathbb{R}^{(C+1) \times d_a}$, and $\mathbf{p}_c$ denotes the prototype of category c . Compared with sample-averaged prototypes, the resulting category prototypes are grounded in the attribute semantic manifold and are therefore more stable under extremely limited supervision.

3.2.2 Visual-Semantic Alignment. With the attribute semantic manifold constructed, we introduce an attribute head to project each visual RoI feature into the semantic space. Specifically, given an RoI feature $\mathbf{f}_i \in \mathbb{R}^{d_f}$, the semantic embedding is defined as

$$\mathbf{e}_i = \text{Norm}(\text{LayerNorm}(\phi(\mathbf{f}_i))) \in \mathbb{R}^{d_a}, \quad (7)$$

where $\phi(\cdot)$ denotes a lightweight MLP and $\text{LayerNorm}(\cdot)$ denotes layer normalization. Before being fed into the attribute head, $\mathbf{f}_i$ is passed through a gradient decoupled layer (GDL) [35] to reduce interference with the backbone parameters.

We then employ a learnable semantic matcher to align visual semantic embeddings with both attributes and category prototypes in the attribute semantic manifold space. Specifically, we define

$$\hat{\mathbf{e}}_i = \text{Norm}(\mathbf{e}_i \mathbf{W}_q), \quad \hat{\mathbf{z}}_n = \text{Norm}(\mathbf{z}'_n \mathbf{W}_k), \quad \hat{\mathbf{p}}_c = \text{Norm}(\mathbf{p}_c \mathbf{W}_k), \quad (8)$$

where $\mathbf{W}_q \in \mathbb{R}^{d_a \times d_a}$ and $\mathbf{W}_k \in \mathbb{R}^{d_a \times d_a}$ are learnable projection matrices, both initialized as identity matrices. This design keeps category and attributes anchored to semantic priors, rather than directly relying on noise-sensitive sample averages.

Let $\mathcal{N}$ denote the set of RoIs with valid category labels, and let $|\mathcal{N}|$ be the number of such RoIs. For each $i \in \mathcal{N}$, let $y_i \in \{1, \dots, C+1\}$ denote its target label, where the $(C+1)$ -th category corresponds to background. Let $\hat{y}_i$ denote the label rectified via the AGR module. We supervise the visual-semantic alignment with three complementary objectives.

First, we directly align each semantic embedding with its corresponding category prototype using a cosine prototype loss:

$$\mathcal{L}_{\text{proto}} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} (1 - \cos(\hat{\mathbf{e}}_i, \hat{\mathbf{p}}_{y_i})). \quad (9)$$

Second, to improve inter-class discrimination in the semantic space, we apply a prototype-based contrastive classification loss:

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \text{CE}\left(\frac{\hat{\mathbf{e}}_i \hat{\mathbf{P}}^\top}{\tau}, \hat{y}_i\right), \quad (10)$$

where $\hat{\mathbf{P}} = [\hat{\mathbf{p}}_1, \dots, \hat{\mathbf{p}}_{C+1}]^\top \in \mathbb{R}^{(C+1) \times d_a}$ is the matrix of projected category prototypes, $\tau \in \mathbb{R}$ is a temperature coefficient, and $\text{CE}(\cdot)$ denotes the standard cross-entropy loss.

Third, we regularize each embedding at the attribute level. Specifically, we compute its similarity to all projected attributes as $s_{i,n} = \hat{\mathbf{e}}_i^\top \hat{\mathbf{z}}_n$, $n \in \{1, \dots, N_a\}$, and obtain the predicted attribute distribution $\pi_{i,n} = \exp(s_{i,n}/\tau) / \sum_{m=1}^{N_a} \exp(s_{i,m}/\tau)$. Let $\boldsymbol{\pi}_i = [\pi_{i,1}, \dots, \pi_{i,N_a}] \in$

$\mathbb{R}^{N_a}$ denote the resulting distribution over attributes. For each category c , we derive a target attribute distribution $\mathbf{q}_c \in \mathbb{R}^{N_a}$ from the corresponding column of the incidence matrix $\mathbf{H}$, with $q_{c,n} = H_{n,c} / \sum_{m=1}^{N_a} H_{m,c}$. The attribute-level alignment loss is then defined as

$$\mathcal{L}_{\text{attr}} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \text{KL}(\mathbf{q}_{\hat{y}_i} \parallel \boldsymbol{\pi}_i). \quad (11)$$

The overall ASM objective is defined as

$$\mathcal{L}_{\text{asm}} = \lambda_p \mathcal{L}_{\text{proto}} + \lambda_c \mathcal{L}_{\text{con}} + \lambda_a \mathcal{L}_{\text{attr}}, \quad (12)$$

where λ_p , λ_c , and λ_a are weighting coefficients for the prototype alignment loss, contrastive classification loss, and attribute-level alignment loss, respectively.

3.3 Attribute-Guided Background Rectification

The AGR module first estimates attribute-guided confidence scores for background RoIs based on the attribute semantic manifold, and then performs confidence-aware supervision rectification to correct biased background supervision.

3.3.1 Attribute-Guided Confidence Estimation. In FSOD, many background RoIs may actually correspond to unlabeled foreground objects, especially from novel categories. These are mistakenly treated as negatives during fine-tuning, leading to incorrect gradients. To alleviate this issue, AGR leverages the early semantic recognition capability of the attribute semantic manifold to estimate attribute-guided confidence scores for background RoIs.

Let $\mathcal{B}$ denote the set of RoIs assigned as background based on ground-truth annotations, which may include unlabeled foreground objects. For each $k \in \mathcal{B}$, we compute its confidence scores over novel categories by measuring the cosine similarity between the aligned semantic embedding $\hat{\mathbf{e}}_k$ and the novel-category prototypes $\hat{\mathbf{p}}_n$, where $n \in \mathcal{C}_{\text{novel}}$:

$$r_{k,n} = \cos(\hat{\mathbf{e}}_k, \hat{\mathbf{p}}_n). \quad (13)$$

To ensure that the identified RoIs are sufficiently discriminative, we further impose a margin criterion. Let $r_k^{(1)}$ and $r_k^{(2)}$ denote the highest and second-highest similarity scores among novel categories, respectively. The confidence margin is defined as $\Delta_k = r_k^{(1)} - r_k^{(2)}$. Only RoIs satisfying $\Delta_k > \delta_m$ are considered for rectification, which filters out ambiguous regions with unstable semantic evidence.

3.3.2 Confidence-Aware Supervision Rectification. Based on the estimated attribute-guided confidence scores, AGR performs confidence-aware supervision rectification over background RoIs. For each $k \in \mathcal{B}$, let $n_k^* = \arg \max_{n \in \mathcal{C}_{\text{novel}}} r_{k,n}$ while $r_k^{(1)} = r_{k,n_k^*}$ denote the corresponding highest confidence score. AGR then applies the following tri-state routing rule:

$$\hat{y}_k = \begin{cases} n_k^*, & \text{if } r_k^{(1)} > \delta_{ps} \text{ and } \Delta_k > \delta_m, \\ -1, & \text{if } \delta_{ps} > r_k^{(1)} > \delta_{bg} \text{ and } \Delta_k > \delta_m, \\ \text{bg}, & \text{otherwise,} \end{cases} \quad (14)$$

where $\delta_{ps} > \delta_{bg}$ denote the pseudo-labeling and masking thresholds, respectively. This rule partitions background RoIs into three cases: (1) high-confidence novel RoIs are assigned pseudo-labels $\hat{y}_k = n_k^*$ and reused as positive samples for online pseudo-labeling; (2) suspicious foreground RoIs are assigned $\hat{y}_k = -1$ and excluded

from background supervision to suppress incorrect negative gradients; and (3) reliable background RoIs retain their original background labels.

Accordingly, AGR rectifies biased background supervision in two complementary ways. RoIs with moderate confidence are masked out from negative supervision, while RoIs with sufficiently high confidence are further reassigned through online pseudo-labeling to recover potential supervision. In this way, AGR mitigates the biased optimization caused by unlabeled foreground objects without introducing additional penalty terms.

3.4 Joint Optimization

During training, the RoI feature is shared by the visual detection branch and the attribute branch. Since the two branches have different optimization objectives, directly back-propagating the attribute loss to the shared backbone may destabilize fine-tuning. To enable semantic supervision while limiting its interference with the backbone, we insert a GDL [35] before the attribute head. Given an RoI feature $\mathbf{f}_i$, GDL produces

$$\hat{\mathbf{f}}_i = \text{GDL}_\lambda(\mathbf{f}_i), \quad (15)$$

whose forward pass is identity, i.e., $\hat{\mathbf{f}}_i = \mathbf{f}_i$, while its backward pass rescales the gradient from the attribute branch as

$$\frac{\partial \mathcal{L}_{\text{asm}}}{\partial \mathbf{f}_i} = \lambda \frac{\partial \mathcal{L}_{\text{asm}}}{\partial \hat{\mathbf{f}}_i}, \quad \lambda \in [0, 1]. \quad (16)$$

That is, GDL leaves the RoI feature unchanged in the forward computation, but attenuates the gradient contributed by the attribute branch when updating the shared backbone. Therefore, the attribute head can still be optimized normally using $\hat{\mathbf{f}}_i$, whereas its influence on the backbone is explicitly controlled by λ .

This follows the core idea of DeFRCN, where GDL preserves the forward feature flow while scaling the backward gradient to decouple upstream and downstream modules. Unlike the original DeFRCN implementation, which applies a learnable affine transformation in the forward pass, our GDL uses an identity forward mapping and only rescales the backward gradient, yielding a lighter decoupling mechanism.

Let $\mathcal{L}_{\text{det}}$ denote the standard detection loss from the visual branch (classification and box regression), and let $\mathcal{L}_{\text{asm}}$ be defined in Eq. (12). The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{asm}}. \quad (17)$$

During novel fine-tuning, AGR is applied to attribute branch labels before computing $\mathcal{L}_{\text{asm}}$, while the visual detection head preserves the standard optimization path. This design keeps the detector training stable and allows lightweight attribute-guided rectification to improve few-shot classification robustness.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. Following standard few-shot object detection protocols [22, 40], we conduct evaluations on two benchmarks: PASCAL VOC [9] and MS COCO [27]. For PASCAL VOC, we utilize the union of VOC 2007 and VOC 2012 trainval sets for training and the VOC 2007 test set for evaluation. The 20 categories are partitioned into 15 base classes and 5 novel classes. Performance is

reported across three standard random splits (Split 1, 2, and 3) for $K \in \{1, 2, 3, 5, 10\}$ shots [22]. We adopt the mean average precision of novel classes at an IoU threshold of 0.5 (nAP50) as the primary metric. For MS COCO, we reserve 20 categories that overlap with PASCAL VOC as novel classes and use the remaining 60 categories as base classes. Training is performed on the train2017 set, while evaluation is conducted on the val2017 set (5,000 images). We report the standard COCO-style AP for $K \in \{1, 2, 3, 5, 10, 30\}$ shots.

4.1.2 Implementation Details. We implement our ASMR method based on MFDC [42] (the baseline method). We use ResNet-101 [20] pre-trained on ImageNet [37] as the backbone. We train our model on $4 \times \text{RTX } 3090$ GPUs with a batch size of 8. For attribute semantic manifold construction, textual attributes generated by ChatGPT are encoded via a pre-trained CLIP (ViT-L/14) text encoder. The number of clusters is set to $N_a = 80$ for PASCAL VOC, and $N_a = 160$ for MS COCO. The embedding dimension is set to $d_a = 768$. To ensure optimization stability, the gradient decoupled layer coefficient λ for the attribute branch is set to 0.01, and the temperature coefficient τ for $\mathcal{L}_{\text{con}}$ and $\mathcal{L}_{\text{attr}}$ is set to 0.07. The weights of each loss are empirically set to $\lambda_p = 1.0$, $\lambda_c = 0.4$, $\lambda_a = 0.5$. For background rectification, the AGR thresholds are empirically set to $\delta_{ps} = 0.8$, $\delta_{bg} = 0.4$, and the discriminative margin is set to $\delta_m = 0.2$. We begin training the attribute branch after 300 iterations as a warm-up strategy to stabilize early fine-tuning.

4.2 Comparison with State-of-the-Art Methods

4.2.1 Results on PASCAL VOC. Table 1 reports the few-shot detection results on the PASCAL VOC benchmark.

ASMR achieves competitive performance across the three novel splits and shows particularly clear advantages in extreme low-shot regimes. This is consistent with our motivation that semantic priors are beneficial when visual supervision is severely limited. In particular, in the 1-shot and 2-shot settings, ASMR consistently outperforms the strong baseline MFDC on most splits. On Novel Split 1, ASMR improves the 1-shot and 2-shot nAP50 from 63.4% to 65.7% and from 66.3% to 67.5%, respectively. On Novel Split 3, the gains are more pronounced, improving the 1-shot and 2-shot performance from 56.1% to 59.6% and from 58.3% to 61.9%, respectively. As the number of annotated instances increases, the performance gap between ASMR and MFDC becomes smaller.

Note that the gains on Novel Split 2 are relatively smaller, and ASMR is slightly below MFDC in several higher-shot settings. A possible reason is that the attributes are obtained through CLIP-based encoding followed by K -means clustering, and the resulting attribute partition may not align equally well with the semantic composition of all category splits. For Split 2, suboptimal clustering of fine-grained attributes may introduce small biases in the assembled category prototypes, which in turn weakens the advantage of semantic priors when visual supervision becomes less scarce.

4.2.2 Results on MS COCO. We further evaluate ASMR on the more challenging MS COCO benchmark to verify its scalability and robustness in complex scenes. The results are given in Table 2.

ASMR achieves the best performance in most shot settings and shows clear advantages in the low-shot regime. In particular, it

Table 1: Few-shot object detection performance (nAP50) on three PASCAL VOC Novel Split sets.

Method / Shot	Novel Split 1					Novel Split 2					Novel Split 3				
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10
FSRW [22]	14.8	15.5	26.7	33.9	47.2	15.7	15.3	22.7	30.1	40.5	21.3	25.6	28.4	42.8	45.9
Meta R-CNN [44]	19.9	25.5	35.0	45.7	51.5	10.4	19.4	29.6	34.8	45.4	14.3	18.2	27.5	41.2	48.1
TFA w/ cos [40]	39.8	36.1	44.7	55.7	56.0	23.5	26.9	34.1	35.1	39.1	30.8	34.8	42.8	49.5	49.8
Retentive R-CNN [11]	42.4	45.8	45.9	53.7	56.1	21.7	27.8	35.2	37.0	40.3	30.2	37.6	43.0	49.7	50.1
QA-FewDet [18]	42.4	51.9	55.7	62.6	63.4	25.9	37.8	46.6	48.9	51.1	35.2	42.9	47.8	54.8	53.5
FSCE [39]	44.2	43.8	51.4	61.9	63.4	27.3	29.5	43.5	44.2	50.2	37.2	41.9	47.5	54.6	58.5
T-GSEL'T [48]	50.4	63.6	61.9	68.6	67.3	31.3	32.9	43.6	47.9	53.9	41.2	49.8	54.1	62.1	61.9
DeFRCN [35]	57.0	58.6	64.3	67.8	67.0	35.8	42.7	51.0	54.5	52.9	52.5	56.6	55.8	60.7	62.5
PS-TTL [15]	58.4	65.7	67.9	69.3	68.1	38.4	47.8	52.8	53.6	49.1	53.0	58.8	59.2	63.8	64.1
AFT [43]	60.1	61.9	63.2	64.3	64.7	43.2	45.7	49.7	48.9	53.9	54.1	57.4	59.2	60.2	63.2
MFDC [42]	63.4	66.3	67.7	69.4	68.1	42.1	46.5	53.4	55.3	53.8	56.1	58.3	59.0	62.2	63.7
ASMR (Ours)	65.7	67.5	67.6	69.4	68.4	43.2	45.5	53.7	53.0	52.6	59.6	61.9	60.8	64.3	65.4

Table 2: Few-shot object detection results on MS COCO (nAP).

Method/Shot	1	2	3	5	10	30
FSRW [22]	–	–	–	–	5.6	9.1
FSCE [39]	–	–	–	–	11.9	16.4
PS-TTL [15]	–	–	–	–	17.3	20.9
T-GSEL'T [48]	–	–	–	–	18.0	22.3
TFA w/cos [40]	3.4	4.6	6.6	8.3	10.0	13.7
QA-FewDet [18]	4.9	7.6	8.4	9.7	11.6	16.5
DeFRCN [35]	6.5	11.8	13.4	15.3	18.6	22.5
AFT [43]	9.6	13.7	15.6	16.8	19.7	23.2
MFDC [42]	10.8	13.9	15.0	16.4	19.4	22.7
ASMR (Ours)	11.4	14.0	15.1	17.0	19.7	21.5

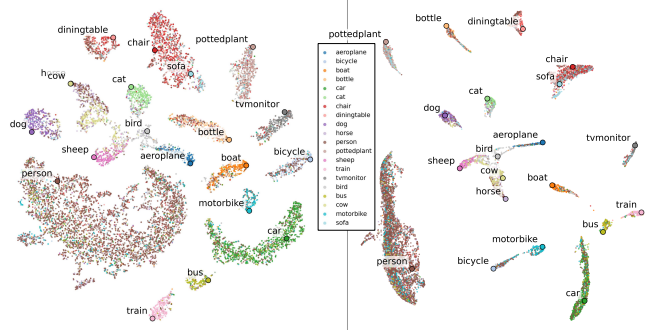
Table 3: Ablation study of the proposed ASM and AGR modules. AGR is decomposed into three rectification components: excluded (excluded suspicious background RoIs), pseudo-labeling (online pseudo-labeling), and margin (confidence margin filtering).

ASM	AGR Components			nAP50 1-shot
	excluded	pseudo-labeling	margin	
				63.40
✓				65.30
✓	✓	✓		65.12
✓	✓		✓	65.21
✓		✓	✓	65.43
✓	✓	✓	✓	65.66

improves over MFDC from 10.8% to 11.4% in the 1-shot setting, and remains consistently better from 2-shot to 10-shot.

4.3 Ablation Study

In this section, we conduct extensive ablation experiments on the PASCAL VOC Novel Split 1 under the 1-shot setting.

**Figure 3: t-SNE (left) and UMAP (right) visualizations of the learned embedding space. Category prototypes $\hat{P}$ and ROI semantic embeddings $\hat{e}_i$ are projected into a shared 2D space.**

4.3.1 Impact of ASM and AGR Components. Table 3 reports the ablation results of the proposed ASM and AGR modules. Starting from the MFDC baseline, introducing ASM alone improves the performance from 63.4% to 65.3% in terms of nAP50, showing that obtaining category prototypes from the attribute semantic manifold provides more robust and transferable representations when visual supervision is extremely limited.

When combining ASM with individual rectification strategies, all variants bring consistent improvements over the baseline, indicating that correcting background supervision is beneficial for FSOD. Specifically, the excluded branch suppresses incorrect gradients from ambiguous background regions, the pseudo-labeling branch recovers potential supervision via high-confidence pseudo-labeling, and the margin constraint filters out semantically uncertain samples. Combining different AGR components leads to further gains, and the full model (ASM + excluded + pseudo-labeling + margin) achieves the best performance of 65.66% nAP50. This result highlights the complementary effects of the three rectification strategies. Overall, the ablation study verifies that ASM mainly improves representation quality, while AGR effectively alleviates negative bias, and their combination yields the most robust performance.

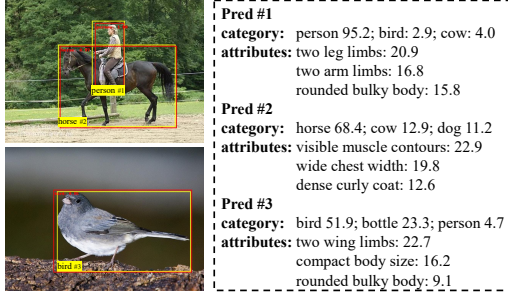


Figure 4: Visualization of category and attribute similarity scores for representative RoIs. For each RoI, we show the top category predictions and the top-3 scored attributes.

4.3.2 Visualization of Semantic Alignment. To further analyze the effect of the proposed attribute semantic manifold, we visualize the learned embedding space using both t-SNE and UMAP. Specifically, we project the category prototypes $\hat{\mathbf{P}}$ and the RoI semantic embeddings $\hat{\mathbf{e}}_i$ from the attribute branch into a shared 2D space. As shown in Fig. 3, category prototypes form structured clusters that reflect shared semantic components across categories, rather than being distributed as isolated points. Semantically related categories (e.g., animals or vehicles) are naturally organized in proximity, indicating that the proposed ASM successfully captures higher-order semantic relationships and imposes a meaningful geometric structure on the embedding space. Moreover, RoI embeddings are consistently distributed around their corresponding category prototypes, demonstrating that the attribute branch effectively aligns visual features with the semantic manifold.

We present both t-SNE (left) and UMAP (right) visualizations for complementary analysis. t-SNE better preserves global neighborhood structures but may introduce distortions in cluster separation, as observed in the partial overlap between similar categories such as horse and cow. In contrast, UMAP provides better local separability, where these categories become more distinguishable. This difference highlights the intrinsic ambiguity of low-dimensional projections rather than deficiencies in the learned representations.

4.3.3 Qualitative Analysis of Attribute Responses. To further examine the effect of attribute learning, we visualize the category and attribute similarity scores for representative RoIs. Specifically, for each RoI, we compute cosine similarities between its semantic embedding $\hat{\mathbf{e}}_i$ and the category prototypes $\hat{\mathbf{P}}$ as well as the attribute prototypes $\hat{\mathbf{z}}_n$, and report the top-ranked categories and top-3 scored attributes. As shown in Fig. 4, the learned RoI embeddings are consistently associated with semantically relevant attributes. For instance, in Pred #1, the RoI is correctly classified as *person*, and its top-ranked attributes, such as “two leg limbs” and “two arm limbs”, correspond well to human body structure.

These examples show that the learned semantic embeddings are not only aligned with category prototypes, but also reflect meaningful attribute-level similarity patterns. This provides qualitative evidence that the proposed attribute semantic manifold successfully guides attribute learning and visual-semantic alignment.

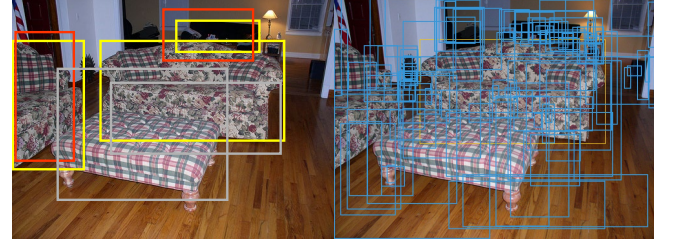


Figure 5: Visualization of AGR. Background RoIs are adaptively rectified into pseudo-labeled foreground (red), excluded regions (gray), and reliable background (blue)

4.3.4 Visualization of Background Rectification. Fig. 5 illustrates the effectiveness of the AGR module on a training image under the 1-shot setting. Although only one sofa instance from the novel category is annotated, the image actually contains three sofa objects. Consequently, many RoIs covering the unlabeled instances are incorrectly assigned as background during training, introducing misleading negative gradients and biasing the detector.

AGR alleviates this issue by exploiting the stable semantic priors provided by ASM. Based on the similarity between each RoI embedding and the semantic prototypes in the attribute manifold, AGR rectifies background supervision into three cases. RoIs with high confidence are converted into pseudo-labeled foreground samples (red), RoIs with ambiguous semantic evidence are excluded from supervision (gray), and RoIs with low affinity to novel categories are retained as reliable background (blue). Only a subset of the reliable background RoIs is visualized, while the red and gray RoIs are selectively shown in the left panel for clarity. This example demonstrates that AGR can effectively distinguish potentially mislabeled foreground regions from true background, thereby suppressing harmful negative supervision and recovering useful training signals even under extreme low-shot conditions.

5 Conclusion

In this paper, we address the key bottleneck of classification discriminability in FSOD, caused by two coupled challenges: semantic drift and negative bias. To tackle these issues, we proposed ASMR, consisting of ASM and AGR modules, which effectively exploits the advantages of the attribute semantic manifold. The ASM module grounds category representations in a structured attribute semantic manifold constructed with LLMs and CLIP, modeling attribute-category relations via a hypergraph and aligning visual RoI features to this manifold to effectively mitigate semantic drift. Based on this manifold, the AGR module identifies potential foreground instances in the background, suppressing incorrect negative gradients and recovering missing supervision through online pseudo-labeling. Extensive experiments on PASCAL VOC and MS COCO benchmarks demonstrate that ASMR achieves competitive performance. Future work will explore the generalization of our method to broader few-shot vision tasks.

References

- [1] Ankan Bansal, Karan Sikka, Gaurav Sharma, Rama Chellappa, and Ajay Divakaran. 2018. Zero-shot object detection. In *Proceedings of the European Conference on Computer Vision*. 384–400.
- [2] Mikhail Belkin, Partha Niyogi, and Vikas Sindhwani. 2006. Manifold regularization: A geometric framework for learning from labeled and unlabeled examples. *Journal of Machine Learning Research* 7, 11 (2006).
- [3] Yoshua Bengio, Aaron Courville, and Pascal Vincent. 2013. Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2013), 1798–1828.
- [4] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in Neural Information Processing Systems* (2020), 1877–1901.
- [5] GuoPing Cao, YuHan Wang, and Lin Chai. 2025. Few-shot object detection integrating multi-level features and attention. In *Proceedings of the 2025 7th Asia Conference on Machine Learning and Computing*. 124–130.
- [6] Yuhang Cao, Jiaqi Wang, Ying Jin, Tong Wu, Kai Chen, Ziwai Liu, and Dahua Lin. 2021. Few-shot object detection via association and discrimination. *Advances in Neural Information Processing Systems* (2021), 16570–16581.
- [7] Riquan Chen, Tianshui Chen, Xiaolu Hui, Hefeng Wu, Guanbin Li, and Liang Lin. 2020. Knowledge graph transfer network for few-shot recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*. 10575–10582.
- [8] Ruoyu Chen, Hua Zhang, Jingzhi Li, Li Liu, Zhen Huang, and Xiaochun Cao. 2025. Generalized semantic contrastive learning via embedding side information for few-shot object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
- [9] Mark Everingham, SM Ali Eslami, Luc Van Gool, Christopher KI Williams, John Winn, and Andrew Zisserman. 2015. The pascal visual object classes challenge: A retrospective. *International Journal of Computer Vision* 111, 1 (2015), 98–136.
- [10] Qi Fan, Wei Zhuo, Chi-Keung Tang, and Yu-Wing Tai. 2020. Few-shot object detection with attention-RPN and multi-relation detector. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4013–4022.
- [11] Zhibo Fan, Yuchen Ma, Zeming Li, and Jian Sun. 2021. Generalized few-shot object detection without forgetting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4527–4536.
- [12] Yifan Feng, Haoxuan You, Zizhao Zhang, Rongrong Ji, and Yue Gao. 2019. Hypergraph neural networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*. 3558–3565.
- [13] Andrea Frome, Greg S Corrado, Jon Shlens, Samy Bengio, Jeff Dean, Marc'Aurelio Ranzato, and Tomas Mikolov. 2013. Devise: A deep visual-semantic embedding model. *Advances in Neural Information Processing Systems* 26 (2013).
- [14] Bin-Bin Gao, Xiaochen Chen, Zhongyi Huang, Congchong Nie, Jun Liu, Jinxian Lai, Guannan Jiang, Xi Wang, and Chengjie Wang. 2022. Decoupling classifier for boosting few-shot object detection and instance segmentation. *Advances in Neural Information Processing Systems* (2022), 18640–18652.
- [15] Yingjie Gao, Yanan Zhang, Ziyue Huang, Nanqing Liu, and Di Huang. 2024. PS-TTL: Prototype-based soft-labels and test-time learning for few-shot object detection. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 8691–8700.
- [16] Amir Globerson, Roi Livni, and Shai Shalev-Shwartz. 2017. Effective semisupervised learning on manifolds. In *Conference on Learning Theory*. PMLR, 978–1003.
- [17] Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, and Yin Cui. 2021. Open-vocabulary object detection via vision and language knowledge distillation. *arXiv preprint arXiv:2104.13921* (2021).
- [18] Guangxing Han, Yicheng He, Shiyuan Huang, Jiawei Ma, and Shih-Fu Chang. 2021. Query adaptive few-shot object detection with heterogeneous graph convolutional networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 3263–3272.
- [19] Guangxing Han and Ser-Nam Lim. 2024. Few-shot object detection with foundation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 28608–28618.
- [20] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.
- [21] KJ Joseph, Salman Khan, Fahad Shahbaz Khan, and Vineeth N Balasubramanian. 2021. Towards open world object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5830–5840.
- [22] Bingyi Kang, Zhuang Liu, Xin Wang, Fisher Yu, Jiashi Feng, and Trevor Darrell. 2019. Few-shot object detection via feature reweighting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- [23] Leonid Karlinsky, Joseph Shtok, Sivan Harary, Eli Schwartz, Amit Aides, Rogerio Feris, Raja Giryes, and Alex M. Bronstein. 2019. RepMet: Representative-based metric learning for classification and few-shot object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [24] Christoph H Lampert, Hannes Nickisch, and Stefan Harmeling. 2013. Attribute-based classification for zero-shot visual object categorization. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2013), 453–465.
- [25] Bohao Li, Boyu Yang, Chang Liu, Feng Liu, Rongrong Ji, and Qixiang Ye. 2021. Beyond max-margin: Class margin equilibrium for few-shot object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7363–7372.
- [26] Yiting Li, Haiyue Zhu, Yu Cheng, Wenxin Wang, Chek Sing Teo, Cheng Xiang, Prahlad Vadakkepat, and Tong Heng Lee. 2021. Few-shot object detection via classification refinement and distractor retreatment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 15395–15403.
- [27] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision*. 740–755.
- [28] Jinlu Liu, Liang Song, and Yongqiang Qin. 2020. Prototype rectification for few-shot learning. In *European Conference on Computer Vision*. 741–756.
- [29] Yen-Cheng Liu, Chih-Yao Ma, Zijian He, Chia-Wen Kuo, Kan Chen, Peizhao Zhang, Bichen Wu, Zsolt Kira, and Peter Vajda. 2021. Unbiased teacher for semi-supervised object detection. *arXiv preprint arXiv:2102.09480* (2021).
- [30] Jiawei Ma, Yulei Niu, Jincheng Xu, Shiyuan Huang, Guangxing Han, and Shih-Fu Chang. 2023. DiGeo: Discriminative geometry-aware learning for generalized few-shot object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3208–3218.
- [31] Anay Majee, Ryan Sharp, and Rishabh Iyer. 2024. SMILe: Leveraging submodular mutual information for robust few-shot object detection. In *European Conference on Computer Vision*. 350–366.
- [32] Sachit Menon and Carl Vondrick. 2022. Visual classification via description from large language models. *arXiv preprint arXiv:2210.07183* (2022).
- [33] Muhammad Ferjad Naem, Yongqin Xian, Federico Tombari, and Zeynep Akata. 2021. Learning graph embeddings for compositional zero-shot learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 953–962.
- [34] Mark Palatucci, Dean Pomerleau, Geoffrey E Hinton, and Tom M Mitchell. 2009. Zero-shot learning with semantic output codes. *Advances in Neural Information Processing Systems* 22 (2009).
- [35] Limeng Qiao, Yuxuan Zhao, Zhiyuan Li, Xi Qiu, Jianan Wu, and Chi Zhang. 2021. DeFCRN: Decoupled faster R-CNN for few-shot object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 8681–8690.
- [36] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. PMLR, 8748–8763.
- [37] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. 2015. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision* 115, 3 (2015), 211–252.
- [38] Christian Simon, Piotr Koniusz, Richard Nock, and Mehrtash Harandi. 2020. Adaptive subspaces for few-shot learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [39] Bo Sun, Banghui Li, Shengcai Cai, Ye Yuan, and Chi Zhang. 2021. FSCE: Few-shot object detection via contrastive proposal encoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7352–7362.
- [40] Xin Wang, Thomas Huang, Joseph Gonzalez, Trevor Darrell, and Fisher Yu. 2020. Frustratingly simple few-shot object detection. In *International Conference on Machine Learning*. PMLR, 9919–9928.
- [41] Jiayi Wu, Songtao Liu, Di Huang, and Yunhong Wang. 2020. Multi-scale positive sample refinement for few-shot object detection. In *European Conference on Computer Vision*. 456–472.
- [42] Shuang Wu, Wenjie Pei, Dianwen Mei, Fanglin Chen, Jiandong Tian, and Guangming Lu. 2022. Multi-faceted distillation of base-novel commonality for few-shot object detection. In *European Conference on Computer Vision*. 578–594.
- [43] Tianxu Wu, Zhimeng Xin, Shiming Chen, Yixiong Zou, and Xinge You. 2025. Adversarial feature training for few-shot object detection. *IEEE Transactions on Circuits and Systems for Video Technology* (2025).
- [44] Xiaopeng Yan, Ziliang Chen, Anni Xu, Xiaoxi Wang, Xiaodan Liang, and Liang Lin. 2019. Meta R-CNN: Towards general solver for instance-level low-shot learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 9577–9586.
- [45] Shuo Yang, Lu Liu, and Min Xu. 2021. Free lunch for few-shot learning: Distribution calibration. *arXiv preprint arXiv:2101.06395* (2021).
- [46] Chenbo Zhang, Bing Huangfu, Hongxu Ma, Jihong Guan, and Shuigeng Zhou. 2025. Multi-modal prototype guided few-shot object detection. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 1852–1861.
- [47] Shan Zhang, Lei Wang, Naila Murray, and Piotr Koniusz. 2022. Kernelized few-shot object detection with efficient integral aggregation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 19207–19216.
- [48] Xu Zhang, Zhe Chen, Jing Zhang, Tongliang Liu, and Dacheng Tao. 2025. Learning general and specific embedding with transformer for few-shot object detection. *International Journal of Computer Vision* 133, 2 (2025), 968–984.

[49] Zhiyuan Zhao, Qingjie Liu, and Yunhong Wang. 2022. Exploring effective knowledge transfer for few-shot object detection. In *Proceedings of the 30th ACM International Conference on Multimedia*. 6831–6839.

[50] Chenchen Zhu, Fangyi Chen, Uzair Ahmed, Zhiqiang Shen, and Marios Savvides. 2021. Semantic relation reasoning for shot-stable few-shot object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8782–8791.